PQCDELAYTEST

Speed-test style frontend sketches

"Test" a network and get its post-quantum migration delay — the way you'd test your internet speed.

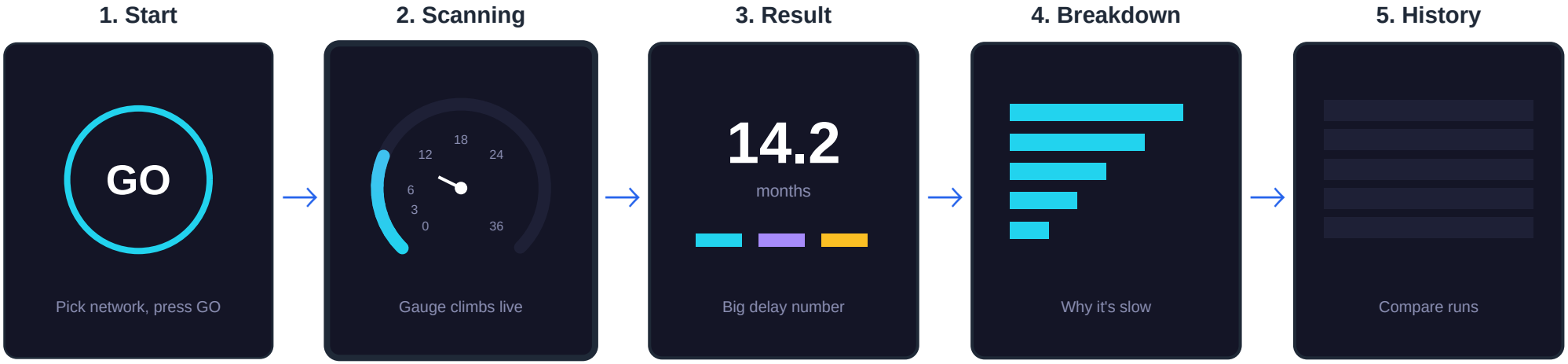
1. User flow overview
2. Start screen (GO button)
3. Live scan with gauge
4. Result screen
5. Detailed breakdown
6. Test history
7. Minimal variant + mobile
8. Synthetic results dataset

All numbers in this document are synthetic placeholders.

1 · Flow

User flow overview

Five screens, one button press. Mirrors the classic speed-test journey: GO → needle moves → big result.



What the "test" actually does behind the scenes

Phase 1: Discover

enumerate hosts, ports & services

Phase 2: Fingerprint

TLS/SSH/IKE handshakes, certs, cipher suites

Phase 3: Benchmark

trial ML-KEM / ML-DSA handshakes → latency & size overhead

Phase 4: Predict

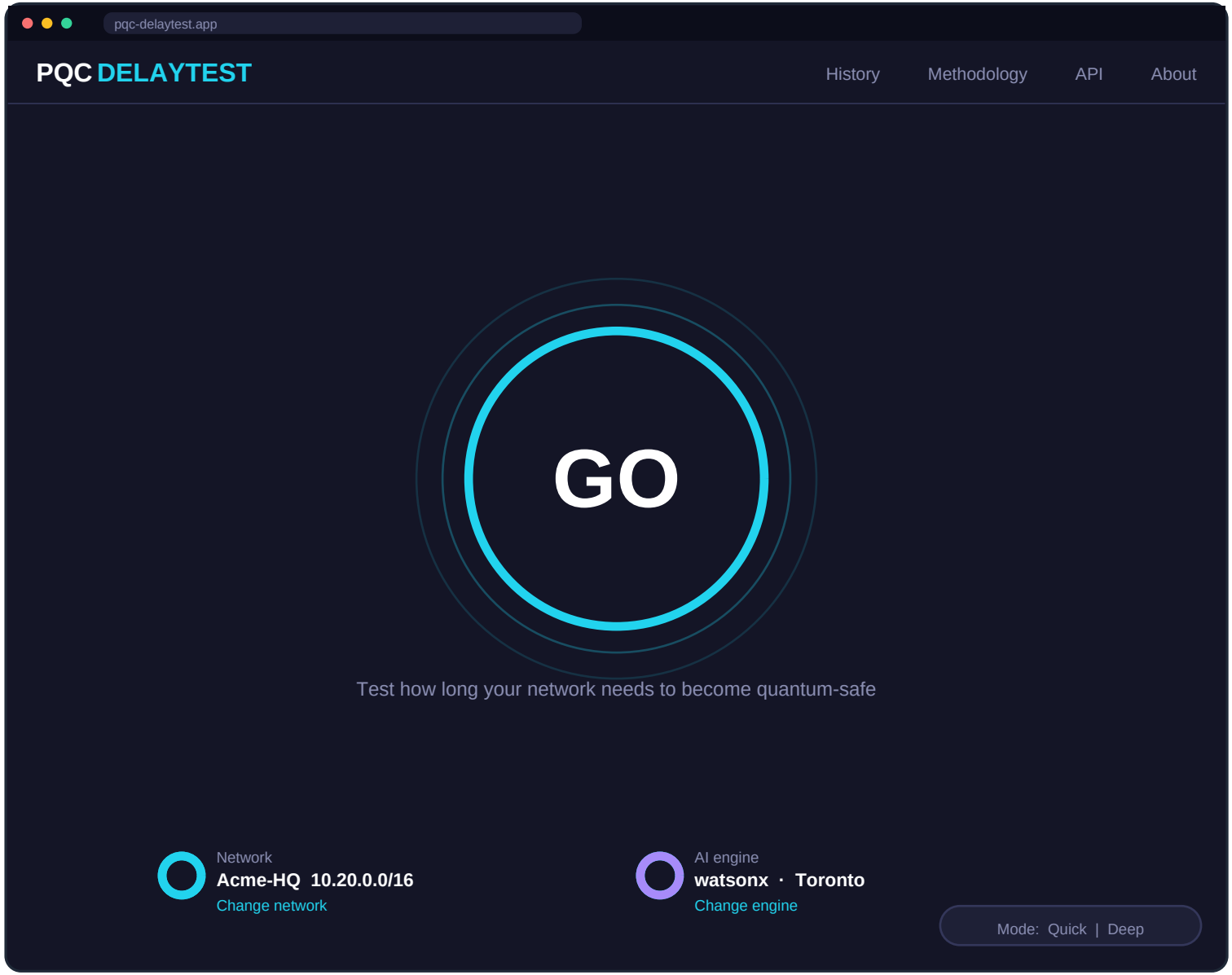
AI model estimates migration delay + confidence

Each phase lights up in turn on the Scanning screen, exactly like download → upload in a speed test.

Screen 1

Start screen

Dark page, one giant GO button, network & AI engine shown underneath — instantly recognisable.



Notes

- Pulsing rings around GO invite the click.
- "Network" replaces the ISP label; "AI engine" replaces the server label.
- Change network opens a small modal: CIDR range, upload inventory CSV, or pick a saved network.
- Quick mode = sample of hosts (~20 s). Deep mode = full scan (minutes).
- Nothing else on the page — that minimalism is the whole speed-test look.

Screen 2

Live scan with gauge

The needle sweeps as the model's running estimate updates; phases tick along the top.

Notes

- Gauge scale is in MONTHS (0–36), not Mbps.
- Needle jitters slightly while the AI refines its estimate, then settles — same drama as a speed test.
- Top row fills in as each phase finishes (like Ping / Download / Upload).
- Sparkline shows the estimate converging.
- Synthetic: needle at 9.5 mo mid-scan, settles at 14.2 mo.

Screen 3

Result screen

Big headline number, secondary metrics in a row, grade badge, share + re-test.

Notes

- Headline uses the same layout as Download speed: label, huge number, small unit.
- Secondary row = Ping/Jitter equivalents: readiness, vulnerable assets, handshake overhead, throughput impact.
- Letter grade A–F makes it shareable ("our network got a C").
- Result ID + Share link let judges open the same result.
- Blocker bar is the bridge to the breakdown page.

Screen 4

Detailed breakdown

Scroll below the result: what contributes to the delay and where the slow parts of the network are.

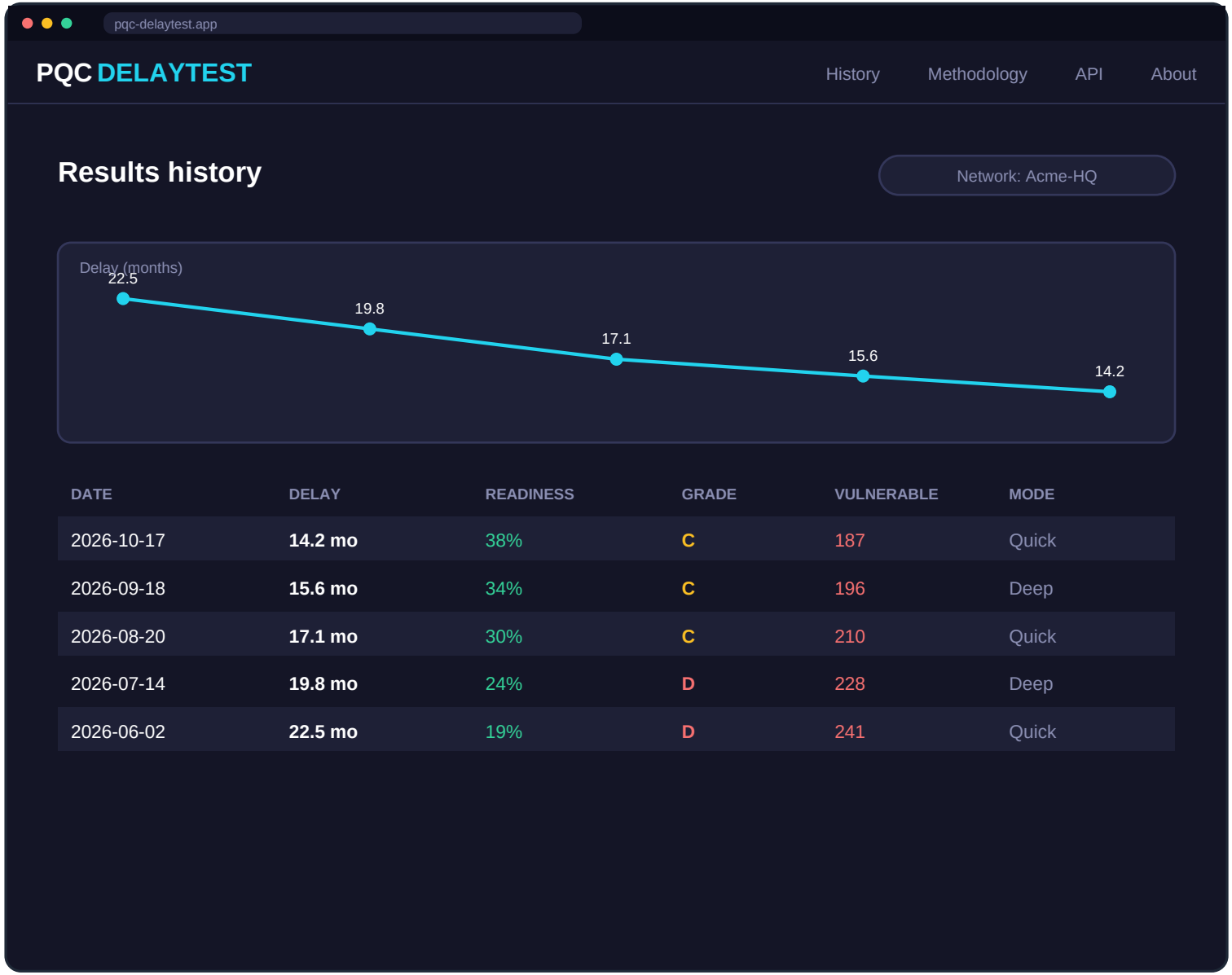
Notes

- Bars rank subsystems by months added — the "why is my test slow" view.
- Algorithm stacked bar shows how much is still classical.
- Performance panel is where the speed-test metaphor is literal: real latency / throughput before vs after PQC.
- Each bar clickable → list of affected hosts.

Screen 5

Test history

Like a speed-test results page: every run listed, so you can watch the delay shrink as you migrate.



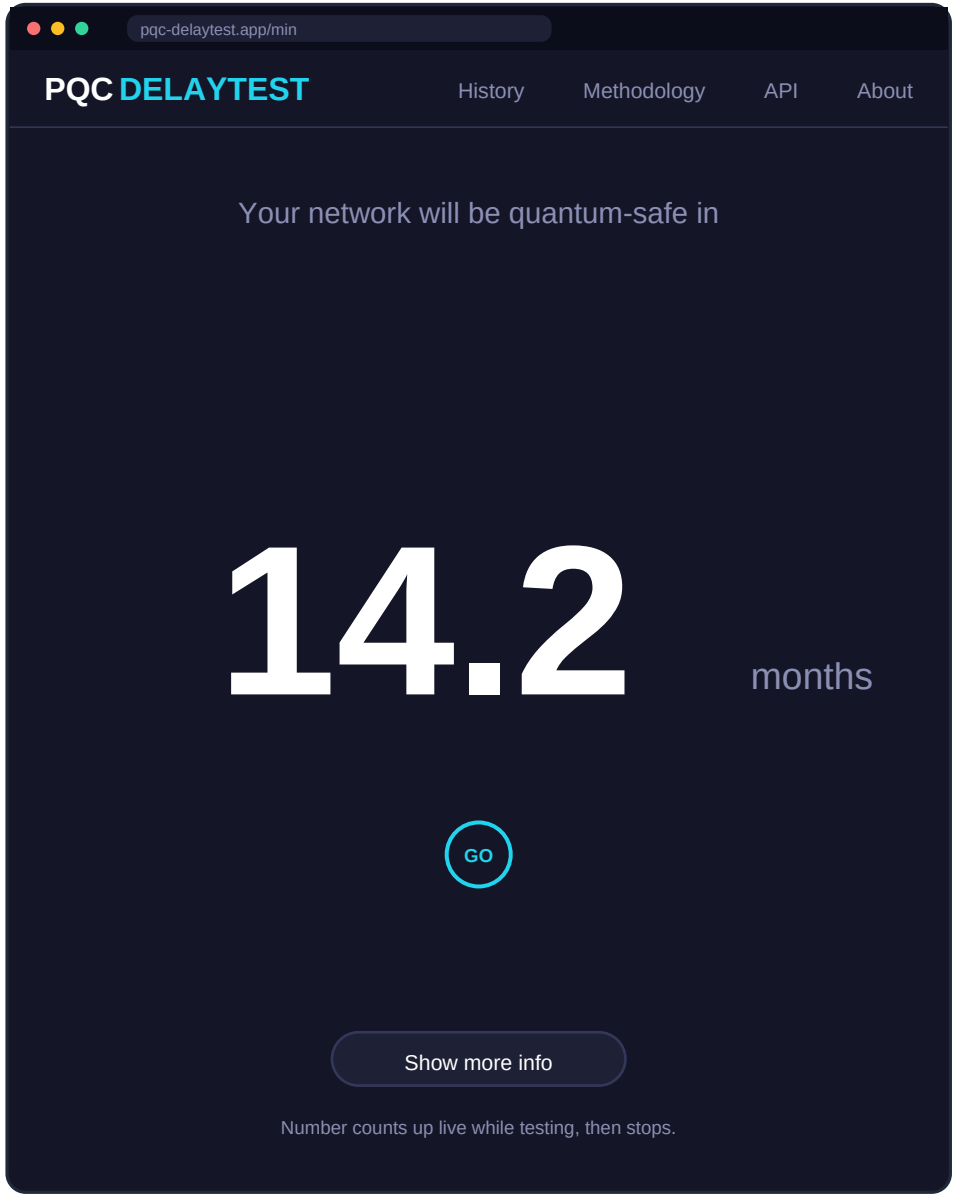
Notes

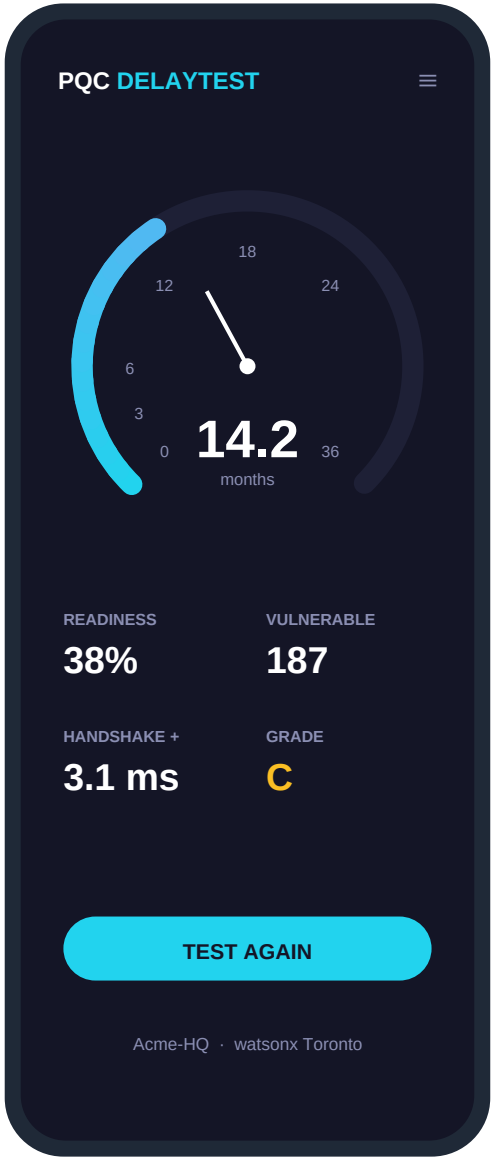
- Trend line proves progress — strong story for a demo: "we went from 22.5 to 14.2 months".
- Rows reuse result ID; click to reopen that result screen.
- Synthetic data: five monthly runs on the same network.
- Export CSV button for the report / judges.

Variants

Minimal variant + mobile

Option A: ultra-minimal single-number page. Option B: the main design squeezed onto a phone.





Notes

- Minimal variant is the fastest to build — good fallback if time runs short.
- Mobile: gauge on top, metrics in a 2×2 grid, one big button at the bottom.
- Both reuse the same API response.

Data

Synthetic results dataset

Mock data the frontend can use before the model is ready. Five example networks + API shape.

Network	Hosts	Delay (mo)	95% range	Readiness	Grade	Vulnerable	Handshake +	Throughput	Top blocker
Acme-HQ (enterprise)	312	**14.2**	11.8–17.0	38%	C	187	3.1 ms	-4.2%	VPN gateway firmware
Northside Hospital	1,480	**27.6**	22.0–34.5	12%	E	1,204	4.8 ms	-6.9%	Medical IoT devices
CloudNative Startup	64	**3.4**	2.6–4.5	81%	A	9	1.2 ms	-1.1%	Third-party SaaS APIs
University Campus	5,920	**21.9**	17.3–28.1	24%	D	3,870	3.9 ms	-5.0%	Legacy SSH/RADIUS
Regional Bank	2,210	**18.3**	15.1–22.4	45%	C	1,056	2.7 ms	-3.6%	HSM replacement cycle

Grade bands: A < 6 mo · B 6–10 · C 10–18 · D 18–24 · E 24–30 · F > 30 months

Suggested API response (POST /api/test → poll GET /api/test/{id})


```
{
  "id": "7F3A-91C2",  "status": "done",  "phase": "predict",  "progress": 1.0,
  "network": { "name": "Acme-HQ", "cidr": "10.20.0.0/16", "hosts": 312 },
  "result": {
    "delay_months": 14.2,  "ci95": [11.8, 17.0],  "grade": "C",  "readiness_pct": 38,
    "vulnerable_assets": 187,  "handshake_overhead_ms": 3.1,  "cert_size_delta_kb": 9.4,  "throughput_delta_pct": -4.2,
    "contributions": [ { "subsystem": "VPN gateways", "count": 31, "months": 5.1 }, ... ],
    "algorithms": { "RSA-2048": 46, "ECDHE-P256": 31, "X25519MLKEM768": 15, "ML-KEM-768": 8 },
    "top_blocker": "31 VPN gateways need firmware with ML-KEM support"
  },
  "live": { "estimate_series": [4, 7, 6, 10, 9, 12, 11, 13, 12, 14, 13.5, 14.5, 14.2] }
}
```


"live.estimate_series" drives the gauge needle and sparkline during the scan; "progress" + "phase" drive the phase pills.